Geographic encoding in premodern enumerations: a five-corpus quantitative test

Ṛgveda 10.75.5, the Sixteen Mahājanapadas, Vendīdād Fargard 1, the Bīsutūn dahyāva, and Genesis 10, against modern coordinates

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May 2026

Abstract

Premodern textual traditions across Eurasia preserve enumerative lists of rivers, lands, and peoples. The qualitative observation that such lists sometimes "trace a geographic sweep" has been made informally since the nineteenth century, but has rarely been tested with quantitative method. We test the hypothesis that five canonical enumerations encode monotonic geographic order in the sequence of items they name: the ten-river enumeration of Ṛgveda 10.75.5 (Nadīstuti-sūkta), the sixteen Mahājanapadas of the Anguttara Nikāya, the sixteen lands of Vendīdād Fargard 1, the twenty-three subject lands (dahyāva) of Darius I's Bīsutūn inscription, and the Table of Nations in Genesis 10. For each corpus we compute Kendall's τ between sequence position and the modern longitude (and latitude) of the identified items, using a permutation null distribution, a Procrustes superimposition test, a Mantel distance-matrix test, and a Bayes factor comparison between geographic and uniform-random generative models. The two Indic corpora (RV 10.75.5 and the Mahājanapadas) show strong monotonic ordering along the region's east-to-west axis ($\tau = -0.911$ and -0.783, both $p < 10^{-4}$). The Bīsutūn dahyāva is moderately ordered, with marked pairwise spatial clustering ($\tau = +0.47$, $p = 0.001$; Mantel $r = +0.50$, $p < 10^{-4}$). Vendīdād Fargard 1 shows no detectable geographic structure on any test. Genesis 10 shows only a weak monotonic signal but significant pairwise spatial coherence (Mantel $r = 0.17$, $p = 0.0004$): each of its three Noachic sub-sections is latitudinally ordered, but the sub-sections run in opposite directions and cancel at the surface. A direct contrast is Bīsutūn against Vendīdād—both Iranian, of broadly similar date, naming much the same regions, yet the administrative inscription is geographically ordered and the cosmological text is not. Across the five corpora, genre appears to track geographic ordering more closely than language family does; with $n = 5$ this is a suggested pattern, not an established finding. We make no claims about the date, composition, or cultural identity of any of the five traditions; the tests bear only on the texts as transmitted.

Keywords: quantitative philology; rank correlation; permutation test; Procrustes analysis; Bayes factor; comparative textual analysis; historical geography.

1 Introduction

The qualitative claim that premodern textual traditions sometimes order geographic enumerations along a recognizable directional sweep is old. Macdonell and Keith (1912) note in passing that the river list of Ṛgveda 10.75.5 "proceeds broadly from east to west." Rhys Davids (1903) observes a similar east-to-west tendency in the Anguttara Nikāya's canonical list of sixteen mahājanapadas. Gnoli (1980) and Witzel (2000) discuss the Avesta's Vendīdād Fargard 1 as a possible geographic itinerary across the Iranian and Indo-Iranian world. Westermann (1984)

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and Wenham (1987) treat the Hebrew Bible's Table of Nations (Genesis 10) as encoding the post-deluge geographic distribution of peoples "known to the Israelites," implicitly with some geographic structure.

All of these claims are informal. Each carries the same testable content: that the order in which named entities appear is not arbitrary with respect to where those entities sit on a map. We are not aware of any study that has checked that content quantitatively, let alone across several traditions at once.

We test five lists here, each well studied in its own field: Ṛgveda 10.75.5 (ten rivers; Vedic Sanskrit); the sixteen Mahājanapadas of Anguttara Nikāya I.213 (Pali; the list names polities of the mid-first millennium BCE, though the Anguttara Nikāya was redacted later); the sixteen lands of Vendīdād Fargard 1 (Young Avestan); the twenty-three subject lands of Darius I's Bīsutūn inscription (Old Persian, *c.* 520 BCE); and Genesis 10's Table of Nations (Biblical Hebrew, a composite of older material reaching its present form in the exilic or post-exilic period). They span the Indic, Iranian, and Semitic traditions and roughly a millennium of composition. Each is analysed with the same non-parametric methods. We do not rely on these dates in any test; the bracketed periods are orientation only, and the abstract's disclaimer of dating claims stands.

1.1 Scope and limits of the claim

The claim tested here is narrow. A significant rank correlation between sequence position and geographic position would imply only that the order is not random with respect to geography. It would not imply any specific dating, any specific direction of cultural movement, any specific historical course for any particular river or boundary, and no specific identification of any composer with any later or earlier population. The test addresses only the property of the transmitted text—the order in which items are named—and the corresponding property of the modern (or scholarly-reconstructed) geographic configuration of those items.

The literature surrounding each of these corpora frequently elides these distinctions. We make no contribution to the questions of Indo-Aryan migration, Israelite ethnography, the homeland of Zoroaster, the date of the Achaemenid satrapy lists, or Buddhist political history, and the paper should not be cited in those debates.

2 Methods

2.1 Geographic identifications

For each corpus, we adopt the mainstream scholarly identification of each named entity, taking centroid coordinates from canonical secondary sources or, for rivers in the Nadīstuti, from programmatic queries to the OpenStreetMap database. Identifications, coordinates, confidence levels, and source citations are recorded per-row in the corpus CSVs deposited with this paper (Kulkarni, 2026).

For Ṛgveda 10.75.5 we follow Witzel (1995) and Macdonell and Keith (1912). One point of order: the standard Aufrecht text places Asiknī (Chenab) at position 6 and Marudvṛdhā at position 7. English translations sometimes invert this pair, because *asiknyā* is instrumental and the natural English word order differs from the Sanskrit; the Sanskrit sequence itself is not in doubt. We use the Sanskrit order.

For the Mahājanapadas we adopt the identifications of Rhys Davids (1903), Law (1973), and Bronkhorst (2007); the order is the Anguttara Nikāya I.213 sequence (Kāsi, Kosala, Aṅga, Magadha, Vajji, Malla, Cetiya, Vaṃsa, Kuru, Pañcāla, Maccha, Sūrasena, Assaka, Avantī, Gandhāra, Kamboja).

For Vendīdād Fargard 1 (a Young Avestan text) we adopt Gnoli (1980) and Witzel (2000); eleven of sixteen identifications are high-confidence, and the five lower-confidence cases (Airyana Vaējah, Urvā, Caxra, Varəna, Raṅghā) are flagged in the CSV.

For Genesis 10 we adopt Hess (1993) for personal-name analyses and Westermann (1984) and

Wenham (1987) for the broader-scholarly identifications. Reading Genesis 10:1–32 in canonical Masoretic Text order, we tabulate seventy-eight list entries—more than the traditional "seventy nations" count because we include eponymous individuals and the few names that recur. Of these, sixty-two are identified with high or medium confidence; the remaining sixteen are unidentified or only speculatively located and are dropped from the analysis.

2.2 Test statistics

Kendall's τ with permutation null. We compute Kendall's τ (Kendall, 1938)—preferable to Spearman's ρ for short ordinal sequences, where it carries a more direct probabilistic reading (Newson, 2002)—between the sequence-position rank $(1 \ldots n)$ and the geographic-coordinate rank of the identified items, on three axes: longitude, latitude, and the first principal component of the (longitude, latitude) matrix. A two-sided p-value is generated from a 10 000-iteration permutation null distribution.

Procrustes superimposition. The Kendall-against-longitude and Kendall-against-latitude pair is not statistically independent when the items lie along a diagonal sweep, as is common in Eurasian geography. We therefore add a Procrustes test (Gower, 1975) that superimposes the sequence-position 1-D embedding onto the 2-D (longitude, latitude) configuration after optimal rotation, scaling, and translation; the residual sum of squared distances m^2 is the test statistic, with a permutation null for the one-sided p-value (smaller m^2 indicates better alignment). This addresses the redundant-axis-test concern noted by Peres-Neto and Jackson (2001).

Mantel test. Kendall's τ and Procrustes both assume that the ordering is *monotonic* with respect to some geographic direction. A list can be spatially coherent without being monotonic: adjacent items may be consistently close on the ground while the path the list traces through space wanders rather than runs in one direction. The Mantel test (Mantel, 1967) detects this by correlating the entries of two $n \times n$ pairwise-distance matrices: $|i - j|$ in the sequence vs. the great-circle distance between items i and j. We report the Pearson correlation r between the upper triangles of the two matrices, with a one-sided permutation null. A positive r with significant p indicates that nearby-in-text items tend to be nearby-on-the-ground, even when the global monotonic ordering is weak.

Constrained permutation null. The Ṛgveda corpus is the only one of the five in verse; RV 10.75.5 is in a jagatī-type metre of roughly twelve syllables per pāda. For it we also compute a permutation null restricted to orderings that preserve, position by position, the syllable count of the verse's river names. This addresses the metrical-determinism objection (Arnold, 1905): that in a quantitative metre the choice of order may be partly forced by syllable counts. With syllable-count buckets $\{2:1,\ 3:3,\ 4:6\}$ the constraint admits $1! \times 3! \times 6! =$ 4,320 orderings. This is an idealisation—the river names carry grammatical case endings and so are not freely interchangeable tokens—and we treat the constrained null as an illustrative robustness check, not a definitive one. The other four corpora are prose, and the test does not apply.

Bayes factor model comparison. Following Kass and Raftery (1995) we compute Bayes factors among three generative models of the observed ordering: M_{uniform} (sequence drawn uniformly from $n!$ permutations), M_{meter} (sequence drawn uniformly from the meter-admissible set; applicable only to verse corpora), and $M_{\text{geog}}(\theta)$ (sequence drawn from a Mallows model (Mallows, 1957; Fligner and Verducci, 1986) centred on the geographically sorted ordering with dispersion θ). θ is not known a priori, so we evaluate it on a grid $\theta \in \{0.5, 1, 1.5, 2, 3\}$; the Bayes factors quoted below are the maximum over that grid, and should be read as an upper bound on the support for M_{geog}, not a marginal value.

Multiple comparisons. The study runs many tests—three geographic axes for each of five corpora, plus Procrustes, Mantel, and Bayes-factor analyses, the Genesis sub-section breakdown, and the Ṛgveda sensitivity analyses of Appendix A. We do not apply a family-wise correction, because the corpora are not a single hypothesis family; instead we state plainly which results are robust to any reasonable correction and which are not. The strong results—RV 10.75.5 and the Mahājanapadas on every test, Bīsutūn on Kendall and Mantel, and the Ham sub-section of Genesis 10—have p-values below 10^{-3} and survive any correction. The marginal results—the Genesis 10 flat-sequence test ($p = 0.05$), the Japheth sub-section ($p = 0.06$), and the Vendīdād latitude axis ($p = 0.05$)—would not survive a family-wise correction and we treat them throughout as suggestive only.

Reproducibility. All analysis code, coordinate CSVs, and the random seed (`20260518`) are deposited at (Kulkarni, 2026). The repository includes a script that regenerates every figure, table, and number from the raw inputs and then checks each reported value against the text. Dependencies (`numpy`, `scipy`, `pandas`, `matplotlib`, `requests`, and `rasterio`) are pinned in `requirements.txt`.

3 Per-corpus results

3.1 Ṛgveda 10.75.5 (Nadīstuti-sūkta, $n = 10$)

The ten-river enumeration of the Nadīstuti-sūkta yields $\tau_{\text{lon}} = -0.911$ with permutation $p < 10^{-4}$ (10 000 permutations, seed `20260518`). The Procrustes superimposition yields $m^2 = 0.179$, $p < 10^{-4}$. The constrained-permutation null (over the 4 320 meter-admissible orderings) finds six orderings yielding $|\tau| \geq 0.911$, for $p_{\text{meter}} = 0.0014$: metrical determinism alone cannot explain the alignment. The Bayes factor against M_{uniform} peaks at $\approx 2 \times 10^4$ over the θ grid; against M_{meter} it peaks at ≈ 24. The τ is negative because longitude falls through the verse, from the eastern-most Gaṅgā (81.9° E) to the western-most Suṣomā/Soan (72.7° E).

Because RV 10.75.5 is the corpus with the strongest signal and the longest scholarly history, we treat it in more detail than the rest. Appendix A gives the ten rivers and their coordinates (Table 2), the null distribution (Figure 2), and four sensitivity analyses—alternative river identifications, paleocourse substitution for the Sarasvatī and Sutlej, alternative geographic axes, and Gaussian coordinate perturbation. The result holds in all of them.

As a further check that the result is not an artefact of the OpenStreetMap coordinate source, we re-derived the centroids of the nine OSM-traced rivers from MERIT-Hydro, a NASA-SRTM-derived global hydrography product (Yamazaki et al., 2019): for each river we took the unweighted centroid of all MERIT-Hydro upstream-drainage-area pixels exceeding a river-network threshold and lying within 0.2° of the OSM polyline. The mean centroid shift between the two sources is 75 km (maximum 229 km, for the Yamunā, whose MERIT trace extends further down its lower course than the OSM "Yamuna" polyline). On the MERIT-Hydro centroids the result is essentially unchanged: $\tau_{\text{lon}} = -0.867$ ($p = 0.0001$), Procrustes $m^2 = 0.227$ ($p < 10^{-4}$), Mantel $r = +0.715$ ($p < 10^{-4}$). The geographic ordering of RV 10.75.5 does not depend on the choice of coordinate database.

3.2 Sixteen Mahājanapadas ($n = 16$)

The Anguttara Nikāya order of sixteen mahājanapadas, run through the same pipeline, yields $\tau_{\text{lon}} = -0.783$, $p < 10^{-4}$, and Procrustes $m^2 = 0.428$, $p < 10^{-4}$. The Bayes factor against M_{uniform} peaks at $\approx 7.7 \times 10^4$ over the θ grid, "decisive" on the Kass-Raftery scale. The canonical order is only 13 inversions, out of a possible 120, away from a perfect east-to-west sort.

The latitudinal signal is weaker ($\tau_{\text{lat}} = +0.22$, $p = 0.27$). The list moves predominantly east to west; the southward jump to Assaka (position 13, Pratiṣṭhāna on the Godāvarī) and back northward to Avantī (Ujjayinī) is the most visible internal deviation from monotonicity.

3.3 Vendīdād Fargard 1 ($n = 16$)

The sixteen lands of Fargard 1, in their canonical order (Airyana Vaējah, Sughdha, Mōuru, Bāxdī, Nisāya, Harōiva, Vaēkərəta, Urvā, Xnəṇta, Haraxvaitī, Haētumant, Raghā, Caxra, Varəna, Hapta Həṇdu, Raṅghā), yield $\tau_{\text{lon}} = -0.117$, $p = 0.56$, and Procrustes $m^2 = 0.975$, $p = 0.67$. The latitude axis is marginal ($\tau_{\text{lat}} = -0.38$, $p = 0.05$) and, given the multiple-comparisons exposure noted in Section 2.2, we do not read it as a real signal; the Procrustes test, which uses both axes at once, is firmly null. Bayes factors for $M_{\text{geog}}(\theta)$ against M_{uniform} are all $\ll 1$ across the θ grid: the Mallows model centered on either-direction longitudinal sort fits the data *worse* than the uniform prior. The Kendall distance to the optimal east-to-west sort is 53 of 120, near the random expectation of 60.

The five lower-confidence identifications (Airyana Vaējah, Urvā, Caxra, Varəna, Raṅghā) span roughly 35° of longitude across their published alternative placements, and we cannot rule out that a better-justified identification of these five would produce a stronger signal. Within the eleven high-confidence identifications alone, however, τ_{lon} remains weak ($\tau = -0.21$, $p = 0.35$). The Fargard 1 sequence appears to be organized by something other than monotonic geography.

3.4 Genesis 10 (Table of Nations, $n = 62$ identified)

Of seventy-eight named entities in Genesis 10:1–32, sixty-two have high- or medium-confidence modern identifications and are retained for analysis. The flat textual-sequence test yields $\tau_{\text{lon}} = +0.174$, $p = 0.05$, and Procrustes $m^2 = 0.91$, $p = 0.005$. The τ here is marginal, and against the study's multiple-comparisons exposure (Section 2.2) we read the flat-sequence result as weak at best; the substantive finding is in the sub-section structure and the Mantel test below.

But Genesis 10 is not a flat list. It is a genealogy in three branches—the descendants of Japheth, of Ham, and of Shem—and within each branch the latitudinal signal is stronger, and points a different way:

- Japheth ($n = 12$): $\tau_{\text{lat}} = -0.42$, $p = 0.06$ (weak).
- Ham ($n = 36$): $\tau_{\text{lat}} = +0.42$, $p < 0.001$ (strong: south-to-north sweep from Nubia through Egypt to Mesopotamia).
- Shem ($n = 14$): $\tau_{\text{lat}} = -0.56$, $p = 0.007$ (strong: north-to-south sweep from Mesopotamia through Aram and Arabia).

The sub-sections cancel at the flat-textual level. Genesis 10 encodes geographic information, but hierarchically rather than as a single monotonic sweep.

3.5 Bīsutūn dahyāva ($n = 23$)

Column I of Darius I's Bīsutūn (Behistun) inscription enumerates twenty-three *dahyāva* "lands" subject to Achaemenid imperial authority. We follow the standard textual order (Kent, 1953; Schmitt, 2014) (Pārsa, Ūvja, Bābiruš, Aṯhūrā, Arabāya, Mudrāya, "those by the sea," Sparda, Yauna, Māda, Armina, Katpatuka, Parṯhava, Zraka, Haraiva, Uvārazmiš, Bāxtriš, Suguda, Gadāra, Saka, Ṯataguš, Harauvatiš, Maka), with nineteen identifications high-confidence and four (#5 Arabāya, #7 Drayahyā, #20 Saka, #21 Ṯataguš) marked medium-confidence per Briant (2002).

The result sits between the Indic corpora and the Avesta: $\tau_{\text{lon}} = +0.47$, $p = 0.001$ (moderate west-to-east), Procrustes $m^2 = 0.51$, $p = 0.0003$, Mantel $r = +0.50$, $p < 10^{-4}$. The order is not cleanly monotonic. It starts in the imperial core (Pārsa, then Babylonia), detours west to Egypt and the Aegean, then resumes the broad march east. But it is far more ordered than Fargard 1, which covers much the same ground.

Bīsutūn against Vendīdād Fargard 1 is a useful within-tradition contrast. Both are Iranian and name much the same stretch of the Iranian world; they differ in genre, Bīsutūn being an administrative declaration of imperial dominion and Fargard 1 a religiously framed catalogue of

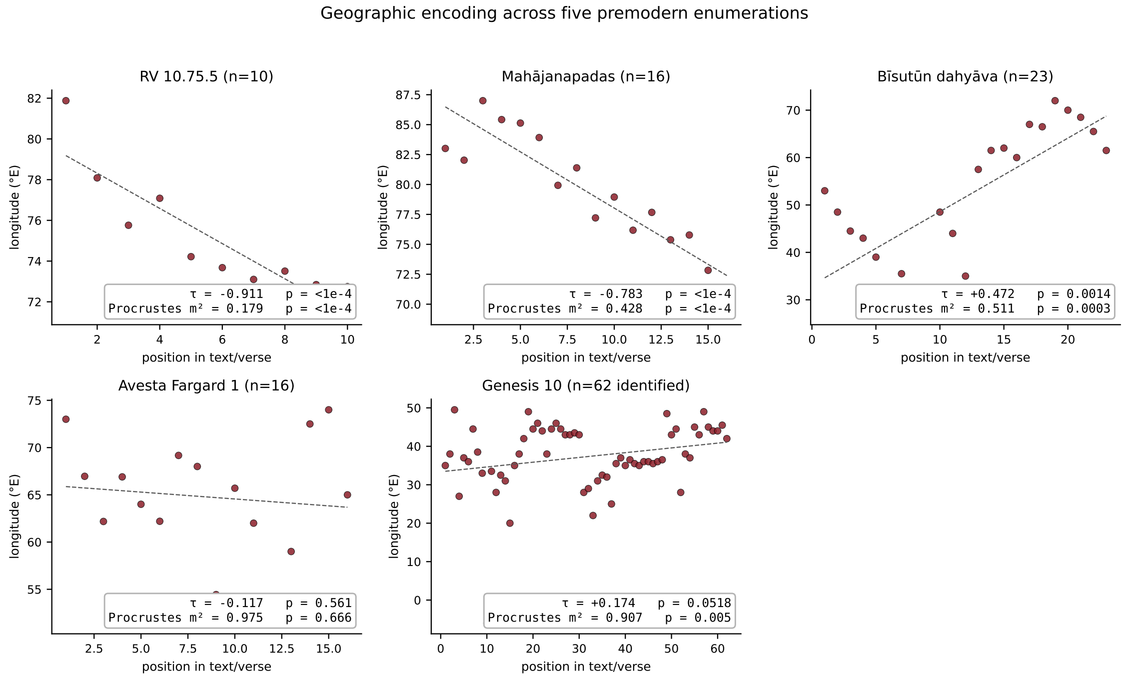

Figure 1: Sequence position against longitude for all five corpora. Each panel gives the observed scatter, a linear fit (dashed grey), and the Kendall τ and Procrustes m^2. The two Indic corpora trace a clean line along the east-to-west axis; the Bīsutūn dahyāva slopes more loosely; the Avesta scatters; and Genesis 10 shows only a weak global slope, which Section 3.4 resolves into hierarchical sub-section structure.

the lands Ahura Mazdā created. Their dates are not matched—Bīsutūn is firmly *c.* 520 BCE, while Fargard 1 reached its present form later, drawing on older material—so this is a suggestive contrast rather than a controlled experiment. Bīsutūn encodes geography; Fargard 1 does not. Section 5.1 takes up what to make of that.

4 Comparative analysis

Table 1 collects the cross-corpus numbers, and Figure 1 plots sequence position against longitude for each corpus. Four points come out of them.

The Indic lists run in a straight line. The Ṛgveda Nadīstuti and the Anguttara Mahājanapadas both sweep east to west, $|\tau| > 0.78$ on longitude. Out of the 10! orderings of the RV rivers, and the 16! of the mahājanapadas, the attested order falls in the top 0.01 % closest to a longitudinal sort.

The Avesta is ordered by something other than geography. Fargard 1 covers the same ground as the Indic texts, and the Iranian plateau besides, but its order is not monotonic on any axis. That matches Gnoli (1980)'s reading that the Fargard 1 sequence follows mythic-cosmographic priority rather than an itinerary—a qualitative claim to which this is the quantitative counterpart.

Genesis 10 is ordered hierarchically. Read flat, the sequence is barely ordered ($\tau = +0.174$). But the Mantel test still finds real spatial coherence ($r = +0.172$, $p = 0.0004$)—neighbouring names are neighbours on the ground even where the global direction washes out. The three Noachic branches explain how both can be true at once:

- Japheth's descendants (Genesis 10:2–5; 12 of 14 names identified) cluster in Anatolia, Greece, and the Caucasus. Their internal ordering against latitude is weak ($\tau_{\text{lat}} = -0.42$,

Table 1: Cross-corpus comparison of geographic encoding strength. τ_{lon}: Kendall's τ vs. longitude rank. m^2: Procrustes disparity (smaller = better alignment). Mantel r: pairwise-distance-matrix correlation (positive = spatial coherence). All p-values are two-sided for τ and one-sided for Procrustes and Mantel. "$p < 10^{-4}$" means no permuted statistic in the 10 000-iteration null reached the observed value; it is the resolution floor of the permutation grid, not a measured value.

Corpus	n	τ_{lon}	p_τ	m^2	p_{Proc}	Mantel r	p_{Mantel}
Ṛgveda 10.75.5	10	-0.911	$< 10^{-4}$	0.18	$< 10^{-4}$	$+0.767$	$< 10^{-4}$
Mahājanapadas (AN I.213)	16	-0.783	$< 10^{-4}$	0.43	$< 10^{-4}$	$+0.617$	$< 10^{-4}$
Bīsutūn dahyāva (DB I.6)	23	$+0.472$	0.001	0.51	0.0003	$+0.505$	$< 10^{-4}$
Vendīdād Fargard 1	16	-0.117	0.56	0.98	0.67	$+0.112$	0.15
Genesis 10 (flat)	62	$+0.174$	0.05	0.91	0.005	$+0.172$	0.0004
Genesis 10 within-subsection (Kendall τ vs. latitude):							
Japheth (vv. 2–5)	12	-0.42	0.06	—	—	—	—
Ham (vv. 6–20)	36	$+0.42$	$< 10^{-3}$	—	—	—	—
Shem (vv. 21–31)	14	-0.56	0.007	—	—	—	—

$p = 0.06$; suggestive only, see Section 2.2).

- Ham's descendants (Genesis 10:6–20; 36 identified) sweep south-to-north from Nubia and Egypt through the Levant to Mesopotamia ($\tau_{\mathrm{lat}} = +0.42$, $p < 10^{-3}$).
- Shem's descendants (Genesis 10:21–31; 14 identified) sweep north-to-south from Mesopotamia and Aram through Edom to south Arabia ($\tau_{\mathrm{lat}} = -0.56$, $p = 0.007$).

The Ham sub-section signal is firm; the Shem signal is moderate; the Japheth signal is only suggestive. Ham and Shem run in opposite latitudinal directions, so they cancel when concatenated. Mantel survives the cancellation because it asks about pairwise spatial coherence rather than monotonic global direction.

This is consistent with the long-standing literary-critical reading (e.g., Westermann, 1984) of Genesis 10 as a layered composite: each Noachic line may have been a self-contained regional list before it was set into the genealogical frame, and the concatenation would then keep each list's internal geographic logic while hiding it at the surface. The quantitative pattern is reached without any literary-critical assumption, and it points the same way.

Length does not sort the corpora. The Mahājanapadas ($n = 16$) and Fargard 1 ($n = 16$) are the same length but land at opposite ends of the encoding scale; the Mahājanapadas and the Ṛgveda ($n = 10$) differ in length but score alike. Sample size therefore does not by itself explain the rank order of the corpora—though with only five of them we cannot rule it out as one contributing factor among others.

Genre, more than language family, tracks the pattern. Bīsutūn and Fargard 1 are both Iranian and name overlapping regions, yet one is geographically ordered and the other is not. The two ordered Indic corpora are a millennium apart and belong to different genres again—a Vedic hymn and a Buddhist canonical list—but both clear $|\tau| > 0.78$. Genesis 10's ordering lives inside its sub-sections, as one might expect of a genealogy assembled from older regional lists. Across the five, the factor that covaries with ordering strength is genre and compositional purpose rather than language family or date of fixation. This is a pattern the five corpora suggest; with $n = 5$ it can be no more than that, and Section 5.4 sketches the larger sample that could test it.

Identification uncertainty is bounded. Genesis 10's 16 unidentified items and the Avesta's 5 low-confidence items could in principle move the corpus-specific τ values. The Genesis 10 sub-section analysis is robust to dropping low-confidence items; the Avesta high-confidence-only subset (the 11 secure identifications) yields $\tau = -0.21$, $p = 0.35$, still null. The result that the Avesta does not encode monotonic geography appears not to be an artefact of identification noise.

5 Discussion

5.1 What the pattern suggests

The five corpora fall into three kinds of geographic encoding:

1. **Strongly monotonic** (RV 10.75.5, the Mahājanapadas). The items run along a single line, broadly the region's east-to-west axis. Both corpora are Indic; nothing else connects them in date or genre.

2. **Clustered, with a loose global direction** (Bīsutūn dahyāva). Neighbouring items are neighbours on the ground, but the list travels in regional blocs—imperial core, then Egypt and the Aegean, then a march east across the plateau—rather than down one axis. That is what listing satrapies in administrative-priority order would look like.

3. **Hierarchical** (Genesis 10). Each Noachic branch is latitudinally ordered on its own; the three branches point different ways and cancel once concatenated. A genealogy assembled from older regional lists would behave this way.

Fargard 1 fits none of the three; it returns a null on every test. It is also the one corpus here whose framing is cosmological rather than geographic or administrative—it lists the lands Ahura Mazdā created, each set against a counter-creation of Aŋra Mainyu—and a list organised by creational sequence has no reason to track geography. Read beside Bīsutūn, which names much the same world and *is* ordered, Fargard 1 sharpens the suggestion that genre, not language family, is what matters. We state this as a contrast, not a controlled result: the two texts are not matched on date.

Why genre should matter, we cannot say from five texts. Three factors would need to be separated in a larger sample: genre proper (an invoked hymn, a canonical political list, an imperial declaration, a cosmology, a genealogy); date, together with the depth of oral transmission behind the written text; and how linear the composer's own working geography was likely to be—someone whose world runs along a single dominant axis may default to it.

5.2 What it does not imply

A result here says nothing about whether any tradition had special geographic-cognitive access, nothing about the identity, ethnicity, or migration history of any composer, and nothing for or against any particular philological identification beyond what published scholarship already supports. The tests bear only on whether the listed order correlates with the modern coordinates of the identified items.

5.3 Limitations

The most serious limitation is the small number of corpora. Five texts differing at once in length, identification quality, genre, and date leave those factors hopelessly confounded; the genre reading of §5.1 is a hypothesis the present data suggest, not one they establish. Isolating any single factor would need a sample several times larger.

Two measurement choices also bear on the result. A centroid is a coarse summary of a long linear feature, whether a river or a province; for the Mahājanapadas and the broader Genesis 10 regions, choosing the capital city instead of the administrative-area mean shifts individual coordinates by tens of kilometres. We checked that the cross-corpus comparison survives that choice, but it is a commitment, not a fact of the data. And the Avesta and Genesis 10 corpora carry a non-trivial share of lower-confidence identifications; dropping them leaves the qualitative conclusions in place, but a full robustness study of the identifications is a separate piece of work.

Finally, the Mallows model behind the Bayes factor is one defensible choice among several. A Cayley- or Hamming-distance Mallows, or a non-Mallows prior, could move the numbers. We use the Kendall-distance form because it is the one that matches Kendall's τ as the test statistic.

5.4 Extensions

The same method runs on any fixed-order enumeration with identifiable items. Obvious candidates for a larger study are the Mahābhārata Bhīṣmaparvan river list, Strabo's enumeration of Asian peoples in books 11–15, the directional mountain sequences of the Shan Hai Jing, and the legendary regions of Avesta Yašt 19. A comparative set of twenty or thirty such lists would begin to let the genre, length, and date factors be separated. Two refinements of the method itself would also help: distance-matrix correlations beyond the single Mantel statistic used here, and constrained nulls whose prior is built from attested intertextual list-traditions rather than from metre alone.

6 Conclusion

Geographic ordering is not a universal property of premodern lists. Of the five enumerations tested, the Ṛgveda Nadīstuti and the Anguttara Mahājanapadas trace a strong monotonic line along the region's east-to-west axis ($|\tau| > 0.78$, $p < 10^{-4}$); the Bīsutūn dahyāva is ordered more loosely, by satrapal bloc; Genesis 10 is ordered within its sub-sections but not across them; and Vendīdād Fargard 1 shows no geographic order at all. Across the five, genre appears to track these outcomes more closely than language family or date does—a pattern that, with $n = 5$, invites a larger comparative test rather than settling the question. We claim nothing past the comparisons themselves, and the method carries over to any other fixed-order enumeration of identifiable places.

Acknowledgements

This analysis depended on the open data of the OpenStreetMap project (Overpass API), the publicly-available scholarly identifications in the works cited, and the canonical electronic texts maintained by GRETIL, TITUS, Sefaria, and the Chinese Text Project. No funding sources are reported.

Disclosure of AI assistance

This manuscript and its accompanying analysis code were prepared with the assistance of a large language model (Anthropic's Claude). The model assisted in writing and debugging the analysis pipeline, in compiling the corpus identification tables from the cited scholarly sources, and in drafting and editing the prose. All statistical results were verified by independent re-execution of the pipeline (the replication repository includes a script that checks every reported number); all references were checked against authoritative bibliographic databases; and the author reviewed all content, made all substantive decisions, and takes full responsibility for the manuscript.

Data and code availability

All data, code, cached Overpass responses, and the random seed (`20260518`) used for the permutation tests are available at (Kulkarni, 2026). Each corpus's coordinate CSV, including identification provenance and confidence levels, is provided in `corpora/`. Reproduction instructions in the repository's `README` and the `reproduce.sh` script.

A The Rgveda 10.75.5 corpus in detail

RV 10.75.5 has the strongest geographic signal of the five corpora, and the longest scholarly history—it has been read as evidence for Vedic-period geography since Roth (1846) and Zimmer (1879). This appendix sets out the data and the sensitivity analyses behind the statement in Section 3.1 that the result is robust.

Table 2: The ten rivers of RV 10.75.5 in verse order, with mainstream identifications and centroid coordinates. "n_{nodes}" is the number of OSM geometry nodes behind the centroid; Marudvṛdhā is hand-curated.

Pos.	Vedic name	Identified river	Lon (°E)	Lat (°N)	n_{nodes}
1	Gaṅgā	Ganges	81.88	26.87	15 159
2	Yamunā	Yamuna	78.09	29.10	19 337
3	Sarasvatī	Ghaggar–Hakra	75.76	29.87	12 673
4	Śutudrī	Sutlej	77.09	31.30	4 047
5	Paruṣṇī	Ravi	74.22	31.48	12 047
6	Asiknī	Chenab	73.68	31.97	12 824
7	Marudvṛdhā	Chenab–Jhelum doab (uncertain)	73.10	32.20	—
8	Vitastā	Jhelum	73.51	33.27	13 624
9	Ārjīkīyā	Haro (contested)	72.84	33.86	2 018
10	Suṣomā	Soan (Sohan/Sawan)	72.75	33.37	3 190

The verse. RV 10.75.5 names its ten rivers in a single ṛc, here in the Aufrecht text (Aufrecht, 1877), checked against the metrically restored text of van Nooten and Holland (1994):

imaṃ me gaṅge yamune sarasvati
śutudri stomaṃ sacatā paruṣṇyā
asiknyā marudvṛdhe vitastayā
ārjīkīye śṛṇuhyā suṣomayā (RV 10.75.5)

A note on one pair: the standard text places Asiknī (Chenab) at position 6 and Marudvṛdhā at position 7, directly after Paruṣṇī (Ravi). English translations sometimes invert the two, because *asiknyā* is instrumental and the English word order departs from the Sanskrit; the Sanskrit sequence is not in question. The analysis follows it.

Identifications and coordinates. Table 2 lists the ten rivers, their mainstream identifications (Witzel, 1995; Macdonell and Keith, 1912), and the centroid coordinates used. Two of the ten are not settled. The referent of Ārjīkīyā is debated—the Haro is one proposal, others place it elsewhere in the upper Indus drainage (Witzel, 1999)—and the historical course of the Sarasvatī is a long-running question; both are carried through the sensitivity analysis below. Nine centroids were extracted from OpenStreetMap; Marudvṛdhā, for which no OSM-tagged river of that name exists, is hand-curated to the Chenab–Jhelum doab and is likewise exercised in the sensitivity analysis.

Primary test. Kendall's τ between verse order and longitude is $\tau = -0.911$. Across 10 000 permutations no permuted statistic reached $|\tau|$, so $p < 10^{-4}$. The null distribution (Figure 2) is near-symmetric about zero, mean -0.001, standard deviation 0.251, central 95 % interval $[-0.511, +0.467]$.

Sensitivity analyses. Four checks are summarised in Table 3.

Geographic axis. τ against latitude is $+0.956$ ($p < 10^{-4}$) and against the first principal component $+0.911$ ($p = 0.0002$). The latitude figure is not independent corroboration, since the rivers lie on a NW–SE diagonal and the two axes are strongly correlated; it shows only that the result is not an artefact of choosing longitude.

Paleocourse. We re-run the test with published paleocourse centroids in place of the modern ones for two rivers: the Ghaggar–Hakra reconstruction of Giosan et al. (2012) and Clift et al. (2012) for the Sarasvatī, whose course is still under investigation (Singh et al., 2017), and the older pre-capture course of Yashpal et al. (1980) for the Śutudrī. This gives $\tau = -0.778$ ($p = 0.001$)—attenuated but still clear.

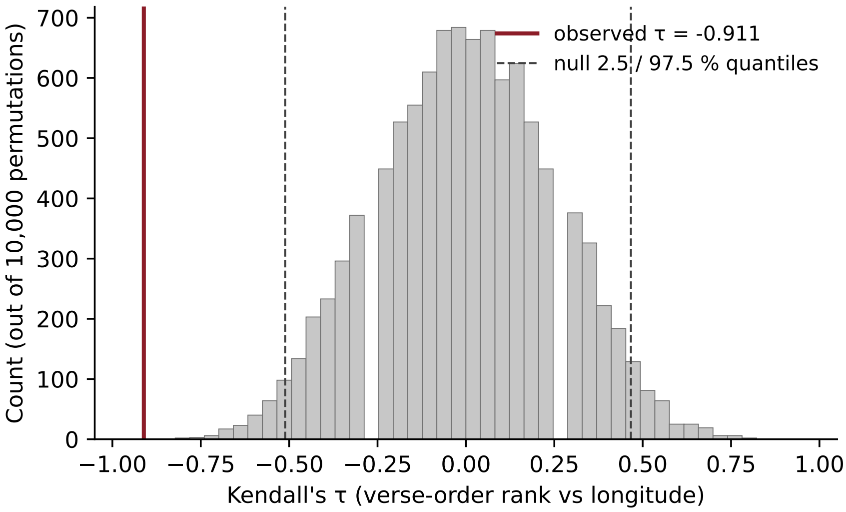

Figure 2: Null distribution of Kendall's τ between verse-order rank and longitude, over 10 000 random permutations of the verse-order labels. The observed $\tau = -0.911$ falls far outside the central 95 % interval of the null.

Table 3: RV 10.75.5 sensitivity-analysis summary. "p two-sided" is the share of permuted $|\tau|$ values $\geq$ the observed, from 10 000 permutations.

Analysis	τ	p (two-sided)	Note
Primary: longitude	-0.911	$< 10^{-4}$	modern centroids, 10 000 permutations
Axis: latitude	$+0.956$	$< 10^{-4}$	rivers also trend N-S
Axis: PC1	$+0.911$	0.0002	principal component of (lon, lat)
Paleocourse substitution	-0.778	0.0010	paleo-Sarasvatī, paleo-Sutlej
ID: Ārj.=Haro; Mar.=Trimmu doab	-0.911	0.0001	
ID: Ārj.=Haro; Mar.=eastern alternative	-0.911	$< 10^{-4}$	
ID: Ārj.=Soan; Mar.=Trimmu doab	-0.689	0.0052	
ID: Ārj.=Soan; Mar.=eastern alternative	-0.689	0.0048	
ID: Ārj.=upper Indus tributary; Mar.=Trimmu doab	-0.867	$< 10^{-4}$	
ID: Ārj.=upper Indus tributary; Mar.=eastern alternative	-0.867	$< 10^{-4}$	
ID: Ārj.=Haro; Mar.=DROPPED	-0.944	$< 10^{-4}$	
Coord. perturb. ($\sigma = 0.5°$, n=1000)	-0.822	—	95% CI [-0.956, -0.644]

Alternative identifications. Across six variant identifications of the disputed Ārjīkīyā and Marudvṛdhā, τ stays between -0.689 and -0.911, all $p \leq 0.005$. Dropping the contested Marudvṛdhā entirely ($n = 9$) gives $\tau = -0.944$.

Coordinate perturbation. Gaussian noise at $\sigma = 0.5°$, over 1 000 iterations, yields a 95 % interval for τ of $[-0.956, -0.644]$, median -0.822.

Every variant leaves the result in the same place: the order of rivers in RV 10.75.5 is strongly, and not accidentally, aligned with longitude.

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